H² Engine V1 Intro Dev Guide

Document section lookups and information

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Chapter I: Creative Clay

When you start a new project, it can feel overwhelming to have such a blank slate, so let's start small and use that as the foundation of your project.

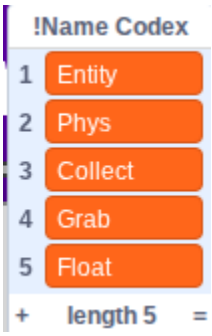
1. Conceptualize the fundamentals

The first step to creating a game is to conceptualize your game properly, or at the very least, the fundamental aspects:

- The gameplay format: puzzle, horror, exploration, etc.
- The stage or game-ending goal: collect all objects, survive for a length of time
- Types of entities you will need.

These will all be used in creating your project, helping you make more informed decisions.

2. Let the engine know what entities you will use



Once you are done conceptualizing what entities you may use in your game, locate the list named ***!Name Codex***. The engine comes pre-installed with 5 entity types, but you may need to make more. Remember that only completely new *types* of entities need a different name code. An important note: your entity name cannot contain periods.

3. Modify your entities' scripts as needed

All entities in the entire game engine are controlled by the ***Entities*** sprite. The engine comes with premade scripts for every entity type; however, you may wish to modify them or add more. All entities use custom blocks that link to their personal behavior types, making it easy to isolate and edit individual entity types.

Chapter II: Mapping the Magic

Now that you have completed the scripts and behaviors of your entities, you can begin to design them and the world around them, including how it looks.

4. Create costumes for your entities

Navigate to the ***Render*** sprite. In the costumes tab, you can modify the premade costumes or create new ones for your entities. Note that our system requires costumes to contain “S” followed by the entity name with the same capitalization as in the ***!Name Codex*** list.

5. Change the map’s wall colors

In the code tab of the ***Render*** sprite, navigate to the section with the heading ***Draw Ray***. Here, you can modify the render properties by adjusting the saturation, brightness, and color blocks within.

6. Design your map

Enter the ***Map*** sprite and create a new costume (or modify the existing one). Make sure to remain at the same scale as the existing map, as our engine does not support vector-mode maps or other scales. We suggest 2px-thick walls and 6px-thick gaps between them.

Chapter III: Saving Success With SEL

After completing the bulk of your game's elements, you may be wondering why your entities seem not to be spawning correctly. This is due to how H² processes "new games." It does this by using our Load system, taking a preset Save-Code string to set up the world. To load your map, player, and entities where you want them, you will need to create a Save-Code that clearly communicates your goals to the engine; however, it isn't as hard as you may think.

7. Open and set up SEL

Open the Scratch project named "SEL Tool" (linked here) and enter the Scratch editor. Run the project. SEL is designed to communicate exactly what our engine does with strings. Before continuing, make sure that the ***!Name Codex Equivalent*** list in SEL completely matches the ***!Name Codex*** list in the main engine.

8. Create your code strings.

Look at the list on the left-hand side of the screen, ***Code Strings***. Here, you will note that each given entry has a command.

1. An entry featuring a word will correspond directly to setting a map.
2. You place down entities by entering their number in ***!Name Codex Equivalent***, followed by x and y coordinates, and if the "entity" or a "phys" type, direction. (Ex., the "Collect" type is 3rd in the list, so starting a Code String with "3." means to place a "Collect"). Note that all data points within an entry are separated by periods.
3. The player, being a special entity, follows the same logic as the entity placement; however, the Player is given the ID of "0" (and, like "Entity"s and "Phys"s, requires a direction attribute)

As you continue to edit, create, and remove lines from the ***Code Strings*** list, you should make sure that ***What H² Engine Would Do*** outputs the correct response. If it does not, you may need to double-check your strings.

9. Bringing it back over

When you have finished editing the strings and the ***What H² Engine Would Do*** properly matches what you want it to do, select all the output in the ***Save Code*** list and copy it. Back in the main engine, locate the ***SAL*** sprite. At the top of the code, locate the ***Load Map From String [string]*** block for the "Start Scene" event. Delete the string in that block, then paste the string from SEL. Make sure to test it by running the engine and clicking *New Game*. The Map, Entities, and Player should all have loaded in correctly. If not, check that you copied the full string from SEL, not just a snippet.

Chapter IV: Game Over

You've successfully completed the hardest part of the development process! Now it's all just the finishing touches. H² Engine does not have preset end conditions or end screens, but you can use easy checks to trigger the ending or other events!

10. Make the win condition

Decide a win condition and find out how to trigger it:

- If you want the player to win by collecting all collectibles, you can track the premade "collectibles collected" variable, and see when it is the desired amount.
- If you want the player to win by reaching a location, create an invisible hitbox that checks for the player.

11. Create the end screen

Make a new sprite for the end screen. Make it hidden by default, but upon meeting the win condition, will show and end the game.

Chapter V: Wrapping up

Now that you have completed main development, let's put in the finishing touches! Open the ***Main Menu*** (or any of the other ***Menu*** sprites) and edit their costumes as you will. Notice how every button has at least two states (whether or not the mouse pointer is currently hovering over them)

And with this, you should be done! Thank you for sharing your story and telling your own tale with the H² Engine developer guide.